

Auteur: Done Goker
Studierichting: Informatica
Oefeningenreeks 1B nr. 24.9.

$$\begin{aligned}1 + \frac{a - bi}{a + bi} + \frac{a + bi}{a - bi} &= 1 + \frac{(a - bi)^2}{a^2 - (bi)^2} + \frac{(a + bi)^2}{a^2 - (bi)^2} \\&= 1 + \frac{(a - bi)^2}{a^2 + b^2} + \frac{(a + bi)^2}{a^2 + b^2} \\&= 1 + \frac{a^2 - b^2 - 2abi}{a^2 + b^2} + \frac{a^2 - b^2 + 2abi}{a^2 + b^2} \\&= \frac{a^2 + b^2 + a^2 - b^2 - 2abi + a^2 - b^2 + 2abi}{a^2 + b^2} \\&= \frac{3a^2 - b^2}{a^2 + b^2}\end{aligned}$$

References